Task 2

Aim

The aim of this experiment is to design and implement a **full-wave bridge rectifier circuit with filter** to rectify sinusoidal and triangular input waveforms. The output captured using a Digital Storage Oscilloscope (DSO) and simulated in Ngspice to compare results.

Equipment and Components used

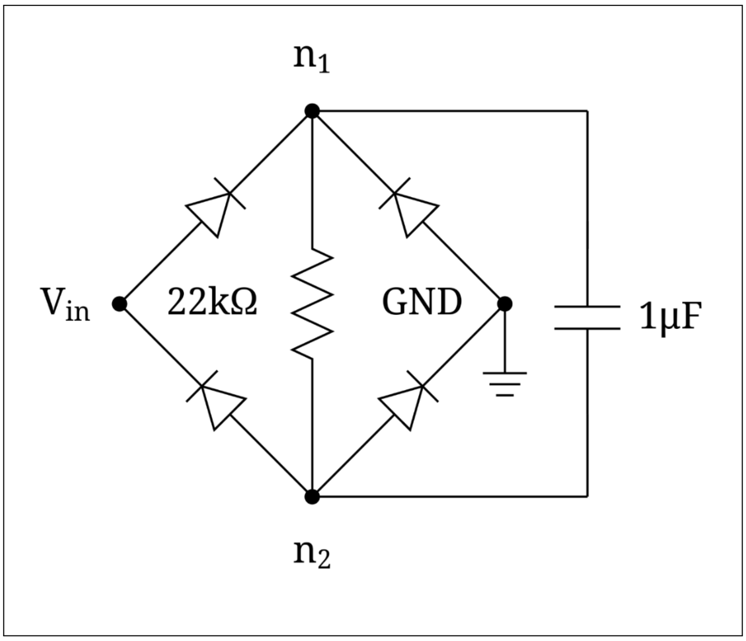
- Fluke 101 CAT-III Multimeter for measuring C and R
- Digital Storage Oscilloscope (Tektronix 1072B)
- Function Generator (Tektronix AFG1022)
- Resistor of 22 kΩ
- Capacitor of 1 μF
- 4 x 1N4007 diode
- Breadboard and jumpers

Procedure

1. Construct a full-wave bridge rectifier circuit using four 1N4007 diodes and connect a filter capacitor across the load with a resistance in parallel to it
2. Configure the function generator in burst mode to provide sinusoidal and triangular wave inputs of 5 V amplitude and $f = 7$ kHz.
3. Set up different cycle lengths ($n = 10, 100$ for sinusoidal and $n = 10, 50$ for triangular inputs).
4. Observe the output waveform on the DSO, using Channel 1 for V_{n1} and Channel 2 for V_{n2} and use *math* function in DSO to find V_{out}
5. Simulate the same circuit in Ngspice and compare the waveforms with experimental results.

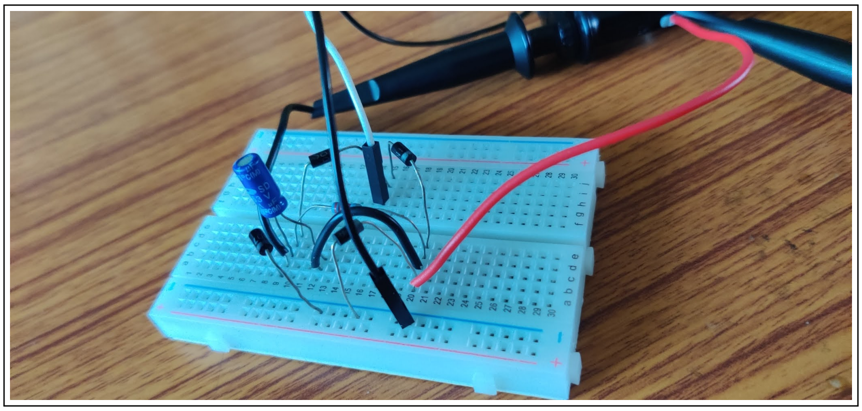
Circuit Design and Setup

The circuit consists of a standard full-wave bridge rectifier made using four diodes. The filter capacitor is connected across the load resistor to smooth out the rectified signal.



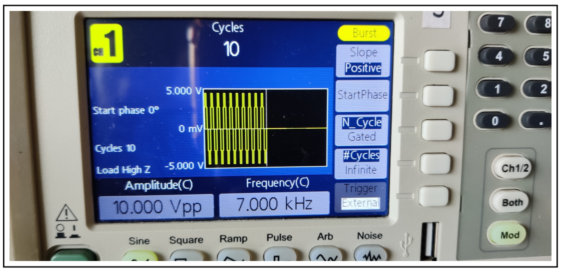
Breadboard implementation of full-wave bridge rectifier with filter

Breadboard Implementation

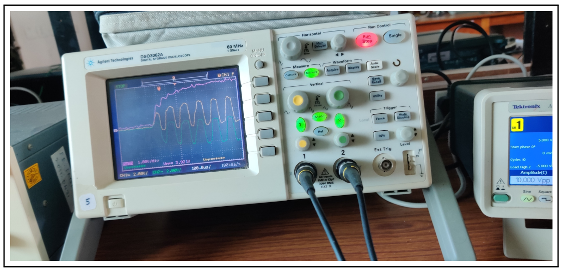


Breadboard implementation of full-wave bridge rectifier with filter

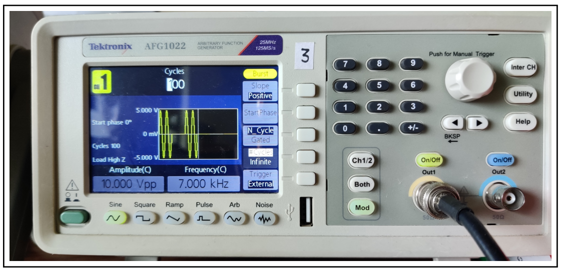
Observations



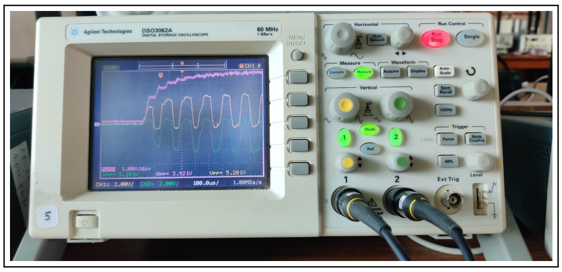
Function generator: Sinusoidal input, $n = 10$



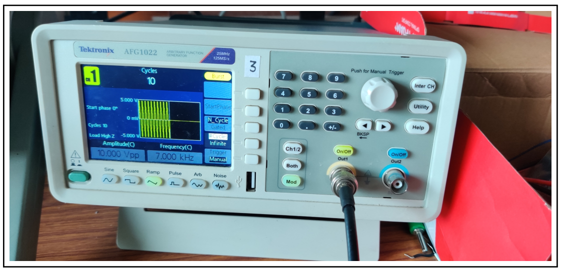
DSO output: Rectified sinusoid, $n = 10$



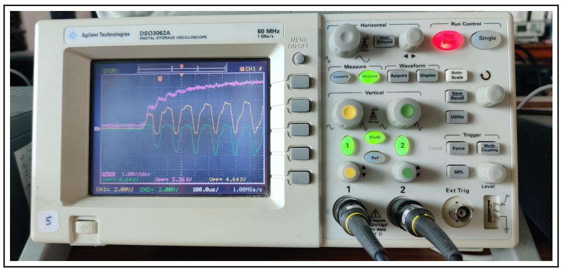
Function generator: Sinusoidal input, $n = 100$



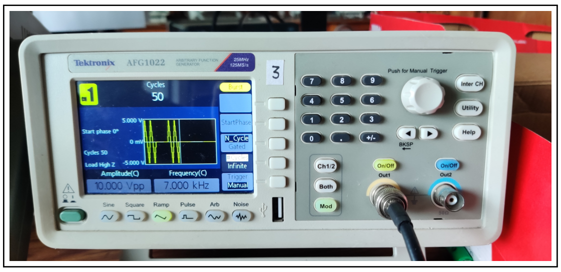
DSO output: Rectified sinusoid, $n = 100$



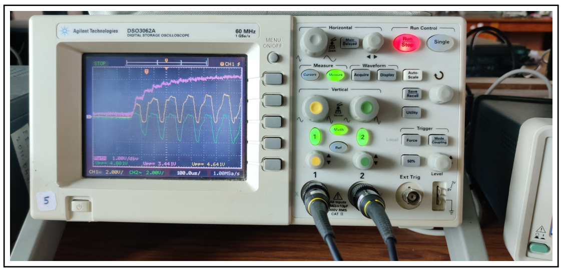
Function generator: Triangular input, $n = 10$



DSO output: Rectified triangular, $n = 10$



Function generator: Triangular input, $n = 50$



DSO output: Rectified triangular, $n = 50$

Simulation using SPICE

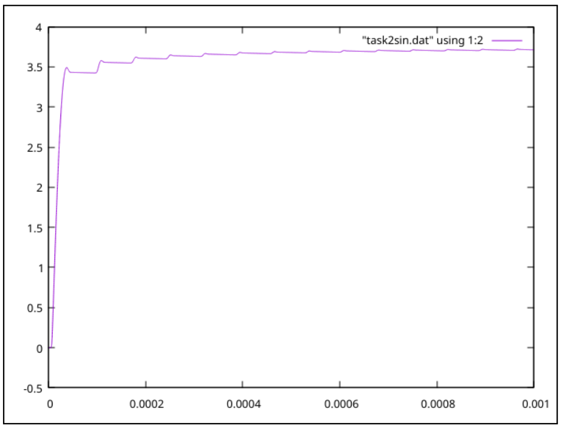
Listing 1: Full Wave Bridge Rectifier Simulation with sinusoid input

```
V1 in 0 SIN(0 5 7000)

D1 in n1 DI_1N4007
D2 0 n1 DI_1N4007
D3 n2 in DI_1N4007
D4 n2 0 DI_1N4007


R1 n1 n2 22k
C1 n1 n2 1u
*C value anything from 1u to 47u is fine
.include 1N4007.mod

.tran 0.1us 1ms
.control
  run
  set wr_singlescale
  wrdata task2sin.dat  v(n1,n2)
.endc

.end
```

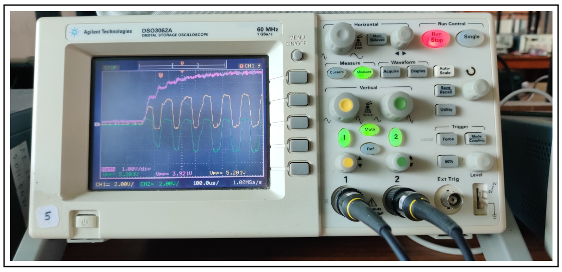
Listing 2: Full Wave Bridge Rectifier Simulation with triangular input

```
BTRI in 0 V = { 5 * (2/pi) * asin( sin( 2*pi*7000*time ) ) }

D1 in n1 DI_1N4007
D2 0 n1 DI_1N4007
D3 n2 in DI_1N4007
D4 n2 0 DI_1N4007


R1 n1 n2 22k
C1 n1 n2 1u
*C value is fine from 1u to 47u

.include 1N4007.mod

.tran 0.1us 1ms
.control
  run
  set wr_singlescale
  wrdata task2tri.dat  v(n1,n2)
.endc

.end
```

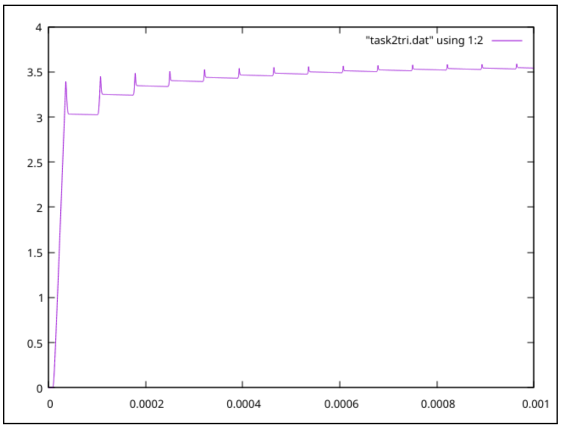
Comparison of Experimental and Simulation Results



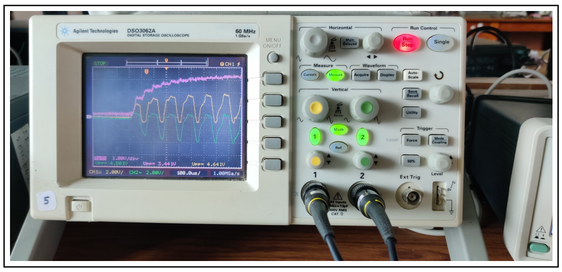
Simulation: Sinusoidal input



Experimental: Sinusoidal input (DSO)



Simulation: Triangular input



Experimental: Triangular input (DSO)

Inference

It was observed that the maximum output voltage (V_{out}) was lower than the expected V_{in}. This is due to the forward voltage drop across the diodes (approximately 0.7 V per conducting diode in the bridge). As two diodes conduct in each half-cycle, the total drop is around 1.4 V, hence $V_{out} < V_{in}$.